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Formation control of a multi-AUVs system based on virtual structure and artificial potential field on SE(3)

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ABSTRACT

This paper presents a formation control method for multiple underwater vehicles that combines artificial potential field and virtual structure. Facing the problem of attitude adjustment in the deployment of AUV and the detection of undulating seabed terrain, a method of finite-time position and attitude tracking control method combined with artificial potential field and virtual structure is presented. We complete the guidance and control in the space of the special Euclidean group of rigid body motions which is SE(3) under the framework of geometric mechanics. The control method we give makes the difference between the trajectory of the AUV and the expected trajectory stabilize to zero in finite time. After that, we give a control method on the attitude, so that the designated pointing vector in the body fixed frame of the AUV converges to the desired direction within a finite-time. This control method of combining artificial potential field and virtual structure can ensure that the AUVs avoid collisions with each other during the dive, and form and change the formation after reaching the set depth. A formation travel plan based on the distance to the seabed for attitude transformation was designed. Simulations are carried out to verify the feasibility.

1. Introduction

With the development of the times, in the military field and marine exploration, underwater unmanned vehicles have increasingly demonstrated their advantages by virtue of their self-propulsion, autonomous control, recycling and repeated use. Compared with a single AUV, the formation coordination system of multiple underwater unmanned vehicles has better characteristics such as higher efficiency, better safety and wider application. The multi-AUVs coordination system can save costs and improve the accuracy of operations, so it has become an inevitable direction of current research (Fernandes et al., 2003; Fiorelli et al., 2006; Wynn et al., 2014).

In the related research in recent years, there are a lot of theses adopting the leader–follower type formation structure. Aiming at the problem of coordinated control of multiple AUVs under the exchange communication topology based on discrete information, a coordinated control protocol with or without time delay is proposed (Yan et al., 2019). The leader–follower type formation structure is used for no time delay. Circumstances, a coordinated control protocol was designed. Some researchers have adopted a fusion control strategy and a redistribution mechanism (RM) based on the virtual structure and leader–follower formation characteristics to improve the speed, stability and accuracy of multiple AUVs. The combination of two path update methods uses self organizing mapping algorithm to enhance the performance of multiple AUVs system (Chen et al., 2020). To eliminate the accumulated errors in the process of dead reckoning, some researchers give a collaborative navigation model and propose a localization approach based on consensus-unscented particle filter algorithm (Xing et al., 2017). Simulations results are provided to verify location performance under the assumption of Gaussian white noise in the systems.

Target search is an important mission of the multi-AUVs system. Some researchers propose a multi-AUVs collaborative target recognition method based on transfer-reinforcement learning (Cai et al., 2020). They transfer learning model based on deep confidence network transfers the feature data of the source domain to the target domain, reducing the repeated calculation of similar data, and thus ensuring the real-time performance of the algorithm. Xia et al. (2021) give a multi-underactuated AUV three-dimensional (3-D) formation control method with a multi-time-scale structure in the presence of uncertain nonlinearities and environmental disturbances. They use the singular

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perturbation method to effectively verify the balance of the exponential stability of the system. For target search and tracking in unknown underwater environment, Cao et al. (2018) show us an integrated algorithm for a cooperative team of multiple autonomous underwater vehicles (Multi-AUVs), which is proposed by combining the Glasius bio-inspired neural network (GBNN) and bio-inspired cascaded tracking control approach to improve search efficiency and reduce tracking errors. In contrast, some researchers establish a potential field function based on the surface water environment information, and then introduced dispersion degree, the homodromous degree and district-difference degree to increase the cooperation of multiple AUV systems, and finally propose an integrated algorithm based on improved potential field (IPF) (Ge et al., 2018). Cao et al. (2015) propose a target search method for multiple underwater vehicles, in which the target attracts AUVs on a global scale through the dynamic neural activity landscape of the model, and obstacles push the AUV away locally to avoid collisions. Finally, the AUV automatically plans its search target path according to the steepest gradient descent rule.

Path planning is also an important research direction of multi-AUVs systems. Wu et al. (2019) present coupled and decoupled multi-autonomous underwater vehicle (AUV) motion planning approaches for maximizing information gain. For multi-Autonomous Underwater Vehicle system task assignment and path planning, Zhu et al. (2018) propose a novel Glasius Bio-inspired Self-Organising Map (GBSOM) neural networks algorithm to solve relevant problems in a Three-Dimensional (3D) grid map.

The virtual structure method (Tanner and Kumar, 2005; Do, 2007) and artificial potential field method (Gazi et al., 2012) have also proved their effectiveness in formation control, so it can be considered to combine these two methods to realize formation control. Compared with the leader–follower type formation structure, the collaborative control method combining artificial potential field and virtual structure has better robustness of the whole system, and it will not be impossible to complete the task because of a certain AUV failure. Moreover, the method of combining artificial potential field and virtual structure has strong scalability, which is convenient to adjust the number of AUVs in the collaborative system according to the task scene. Compared with other methods, the artificial potential field and virtual structure method are easy to set parameters to complete the obstacle avoidance between AUVs, which makes the conditions for deploying AUVs less stringent. Therefore, the formation control method combining artificial potential field and virtual structure is suitable for controlling the attitude of each vehicle in formation mission to achieve better mission detection effect.

The seabed is an important lower boundary of the underwater sound field, and its acoustic characteristics and the fluctuation of the seabed topography are extremely important to the sound propagation. Some papers (Urick, 1954; Chang et al., 2014; Wen et al., 2015; Dai et al., 2021) pointed out that rough seabed would have an impact on acoustic communication, matching field localization, and seabed parameter inversion. For example, topographic changes such as seamounts and ocean slopes would have an impact on underwater sound propagation. Therefore, it is of great significance to maintain a certain distance between the seabed to obtain better sonar data when the underwater vehicle cooperative system detects the seabed. Facing the undulating environment of the seabed, in order to make the underwater acoustic detection and optical detection results better, it is necessary not only to synchronize the trajectory of the AUV, but also to quickly converge its attitude information to the desired attitude. In order to unify the target tracking algorithm to facilitate and debug, a finite-time tracking and attitude tracking control method based on SE(3) is adopted here. Similar results using the geometric framework have been applied to the stabilization and tracking of the orbiting spacecraft's attitude and angular velocity under gravity (Chaturvedi and McClamroch, 2006; Sanyal and Chaturvedi, 2008). The use of similar methods for AUV control has also been mentioned in Woolsey (2006) and Singh et al. (2009).

Table 1
The definition of variables.

Symbol	Description
$m \in \mathbb{R}$	The mass
$J \in \mathbb{R}^{3*3}$	The inertia matrix with respect to the body-fixed frame
$R \in SO(3)$	The rotation matrix from the body-fixed frame to the inertial frame
$\Omega \in \mathbb{R}^3$	The angular velocity in the body-fixed frame
$x \in \mathbb{R}^3$	The location of the center of mass in the inertial frame
$v \in \mathbb{R}^3$	The velocity of the center of mass in the inertial frame
$f \in \mathbb{R}$	The total thrust
$\tau \in \mathbb{R}$	The thrust generated by the propeller along the b_1 axis
$M \in \mathbb{R}^3$	The total moment in the body-fixed frame

We generate a set of waypoints by combining the artificial potential field and virtual structure introduced, and the waypoints are given in the form of tuples. And the tuple is composed of the position vector of the vehicle and the vehicle's attitude at that position. We can get the two differentiable expected position and pointing direction trajectory which are parameterized by time from the waypoint. We use the controller to make the position vector converge to the desired position in a finite-time. After that, we generate the required position information according to the direction of the position control vector and the required pointing direction. In this way, an almost global finite-time stabilization attitude tracking control method can be used to track pre-generated path and attitude information.

The dynamics of a AUV is expressed globally on the configuration manifold of the special Euclidean group SE(3). We construct a tracking controller to follow prescribed trajectories for the center of mass and heading direction. It is shown that this controller exhibits almost global exponential attractiveness to the zero equilibrium of tracking errors. Since this is a coordinate free control approach, it completely avoids singularities and complexities that arise when using local coordinates.

The rest of the paper is organized as follows: Section 2 will introduce some useful notations and preliminaries knowledge and the dynamics model of AUV on SE(3). Section 3 states the formation design of multi-AUVs which is applied with the combination of virtual structure and artificial potential field. Section 4 explains the trajectory generation and error dynamics of AUVs during navigation. Section 5 gives the designed AUV control strategy on SE(3). The simulation results and conclusions of the designed two cases are given in Section 6 and Section 7, respectively.

2. Background and model description

2.1. Notations

The variables related to the theoretical and mathematical representation of this paper are defined in Table 1.

2.2. Dynamics model

The configuration space of the AUV modeled as a rigid body is defined by the location of the center of mass and the attitude with respect to the inertial frame. $x \in \mathbb{R}^3$ represents the position vector of the origin of the vehicle body frame $\mathcal{B}$ with respect to the inertial frame $\mathcal{I}$ shown in the frame $\mathcal{I}$. Let $R \in SO(3)$ denote the direction (attitude), defined as the rotation matrix from the frame $\mathcal{B}$ to the frame $\mathcal{I}$. Then the attitude and position information of the entire AUV can be expressed as a matrix:

$$g = \begin{bmatrix} R & x \\ 0 & 1 \end{bmatrix} \in SE(3) \tag{1}$$

where $SE(3)$ is the six-dimensional rigid body motion Lie group, which is obtained as the semi-direct product of $\mathbb{R}^3$ and $SO(3)$. $x^d(t)$ gives the ideal trajectory on $\mathbb{R}^3$, where $x^d(t)$ is continuous and can be

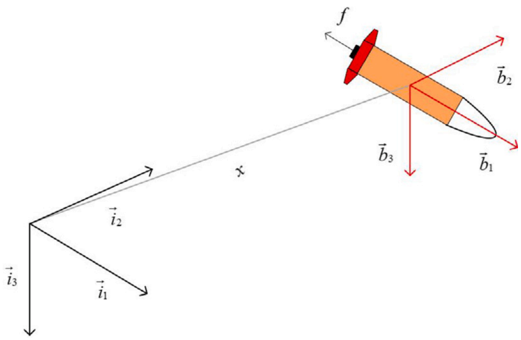

Fig. 1. The inertial reference frame and body fixed frame.

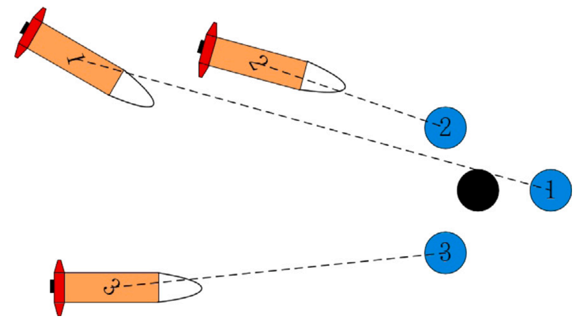

Fig. 2. Schematic diagram of multi-AUVs formation system. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

differentiated twice. Here, the virtual structure method with artificial potential field will be used to generate the ideal trajectory, and the collision avoidance between the AUV will be completed at the same time.

We choose an inertial reference frame $\vec{i}_1, \vec{i}_2, \vec{i}_3$ and a body fixed frame $\vec{b}_1, \vec{b}_2, \vec{b}_3$. The origin of the body mount is located at the center of mass of the vehicle. The first and second shafts of the body mount $\vec{b}_1, \vec{b}_2$, as shown in Fig. 1. The third body fixed axis $\vec{b}_3$ is perpendicular to the plane. The direction of $\vec{b}_1$ is opposite to the direction of the total thrust.

We assume that the thrust of propeller is directly controlled, and the direction of thrust is perpendicular to the plane formed by $\vec{b}_2$ and $\vec{b}_3$. The total thrust is f, acting in the direction of $-\vec{b}_1$. According to the definition of the rotation matrix $R \in SO(3)$, the direction of the fixed axis $\vec{b}_i$ of the ith object is given by Re_i in the inertial system, where $e_1 = [1;0;0], e_2 = [0;1;0], e_3 = [0;0;1] \in \mathbb{R}^3$. The total thrust is given by $-fRe_1 \in \mathbb{R}^3$ in the inertial frame.

We assume that the geometric center of the AUV and the center of gravity are almost coincident, so that we can ignore the torque generated by the resistance. We indicate that the kinematic equations of the AUV are

$$\dot{x} = Rv \tag{2}$$

$$\dot{R} = R\hat{\Omega} \tag{3}$$

the $hatmap\hat{\cdot} : \mathbb{R}^3 \to SO(3)$ is given by

$$\hat{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix} \in SO(3) \tag{4}$$

We assume that the AUV is immersed in the viscous fluid, and the origin of the body-fixed frame of the body is at the center of gravity. The total kinetic energy of the system composed of rigid bodies and fluids is given by

$$T = \frac{1}{2}\begin{pmatrix} v \\ \Omega \end{pmatrix}^T \begin{pmatrix} mI + M_f & O \\ 0 & J_b + J_f \end{pmatrix} \begin{pmatrix} v \\ \Omega \end{pmatrix} \tag{5}$$

where I is the 3×3 identity matrix, M_f is the added mass, J_b and J_f are body inertia matrix and added moment of inertia respectively (see more hydrodynamics text in Fossen (2011)).

The dynamics which evolves on TSE(3) are trivialized to SE(3) $\times\mathfrak{se}(3)$. We assume here that the submerged weight of the AUV is equal to the buoyancy force, and then we can get the dynamics equations of the AUV on SE(3) $\times\mathfrak{se}(3)$ (see more details in Sanyal et al. (2010))

$$M\dot{v} = Mv \times \Omega + D_v(v)v - fRe_1 \tag{6}$$

$$J\dot{\Omega} = J\Omega \times \Omega + Mv \times v + D_\Omega(\Omega)\Omega + \tau \tag{7}$$

where $M = mI + M_f$, $J = J_b + J_f$.

3. Multi-AUVs formation design

The black dot represent the formation reference point, and the surrounding light dots represent the expected virtual structure; the blue dots represent the virtual structure mass points, as illustrated in Fig. 2. The formation reference point is used as the center of the formation, and the desired virtual structure formation is formed around it as the desired position of the virtual structure mass point. At the initial moment, multiple AUVs are randomly distributed within a certain range when they are deployed, and the virtual structure particles are at the same initial position as the AUV and move to the desired virtual structure. During the movement, the artificial potential field is set with the virtual structural mass point as the center to realize collision avoidance and obstacle avoidance. The AUV uses virtual structural particle as the tracking target. After reaching the desired depth, under the control of the target tracking algorithm, it finally realizes its own formation to explore the underwater terrain (see more in Leonard and Fiorelli (2001)).

For the ith AUV, the control objective on the kinematics level is to converge the position of the AUV to the desired formation position by given the desired attitude and desired speed of the AUV, so that the multi-AUVs system can form a stable formation, and in this process, avoid collisions with each other and maintain the necessary safety distance.

The virtual structure mass point model is

$$\begin{cases} \dot{p}_i = \mathfrak{u}_i, \\ \dot{\mathfrak{u}}_i = \varrho_i, \end{cases} \tag{8}$$

where $p_i \in \mathbb{R}^3$, $\mathfrak{u}_i \in \mathbb{R}^3$ are the position and velocity of the ith particle in the inertial frame; $\varrho_i \in \mathbb{R}^3$ are the control input of the ith particle.

$$\varrho_i = c_1(p_{ri} - p_i) + c_2(\mathfrak{u}_{ri} - \mathfrak{u}_i) + F_i \tag{9}$$

where c_1 is the position error feedback coefficient; c_2 is the speed error feedback coefficient; $p_{ri} \in \mathbb{R}^3$ and $\mathfrak{u}_{ri} \in \mathbb{R}^3$ are respectively the position and speed of the ith mass point corresponding to the expected virtual structure in the inertial frame; $F_i = -\nabla_{qi}\mathfrak{V}(q)$ is the negative gradient term of the artificial potential field to realize collision avoidance between particles.

Simplify F_i to 0, the simplified control input is

$$\varrho_i = c_1(p_{ri} - p_i) + c_2(\mathfrak{u}_{ri} - \mathfrak{u}_i) \tag{10}$$

Substitute Eq. (10) into Eq. (8), we get

$$\ddot{p}_i + c_2\dot{p}_i + c_1 p_i = c_2\dot{p}_{ri} + c_1 p_{ri} \tag{11}$$

It appears to be a linear second-order system. To make the system stable, we should make $c_1 > 0$ and $c_2 < 0$ according to the *Hurwite* criterion.

As the tracking target of the AUV, the virtual structure particle should avoid strenuous movement to reduce the frequent maneuver of the AUV. Choose c_1 and c_2 to make this second-order system an over-damped system, which can make c_1 smaller and c_2 larger. Since the position of the AUV at the initial moment is randomly distributed within a certain range, the position of individual AUVs may be far away from the formation reference point. A smaller c_1 can prevent the virtual structure particles from moving to the formation reference point too fast, which will saturate the AUV speed; if c_2 is larger, the AUV can be matched in speed to facilitate the stable formation of the formation.

In the discussion on the control input of the virtual navigator in the previous section, the discussion about the collision avoidance force item was ignored. In the actual formation process, the formation members may collide with each other during the movement. Therefore, it is necessary to study the collision avoidance problem between AUVs. In the previous chapters, the virtual navigator is used as the tracking target of the AUV to realize the formation of the formation. In the research on collision avoidance, the virtual navigator is still the research object. The artificial potential field used for collision avoidance is applied to the virtual navigator. When the virtual navigator realizes collision avoidance under the effect of the collision avoidance algorithm, the AUV indirectly realizes mutual collision avoidance by tracking the corresponding virtual navigator. The artificial potential field used in this chapter only produces repulsive force in a small area, and has no force in other areas. The potential field force is only used for collision avoidance, so the design of the artificial potential field can be simplified.

Considering that the particle i is in the potential field generated by other particles nearby, the total potential energy of the ith particle can be expressed as

$$\mathfrak{V}(p) = \sum_{j,i=1}^{N} \zeta(\|p_j - p_i\|), j\neq i \tag{12}$$

$\varphi(x)$ is the potential energy function, and $\|\cdot\|$ is the modulus of the vector. That is, the artificial potential field energy is a function of the relative distance of the particles. Since only considering the artificial potential field for collision avoidance, rather than organizing the formation, only the potential field is required to have repulsive force. Construct potential function

$$\zeta(z) = \begin{cases} \frac{1}{2}\mathbf{k}(\frac{1}{z} - \frac{1}{\mathcal{R}}), & 0 < z \leq \mathcal{R} \\ 0, & z > \mathcal{R} \end{cases} \tag{13}$$

$\mathbf{k}$ is the potential field strength coefficient, and its selection should consider the inertia of the AUV and the control output of the actuator; if $\mathbf{k}$ is too small, it is not conducive to safe collision avoidance; if $\mathbf{k}$ is too large, it may exceed the output range of the actuator which has no practical meaning; $\mathcal{R}$ is the range of potential field.

In order to avoid the interference of the artificial potential function on the formation, the distance between the particles in the final formation should be greater than the range of the potential function, namely

$$\|p_j - p_i\| \geq \mathcal{R} \qquad \forall i, j \in N, j \neq i \tag{14}$$

The artificial potential field effect in the formation process can be regarded as interference to the second-order system. The existence of the negative gradient term of the artificial potential field causes the particle to move in the direction where the energy of the artificial potential field is reduced, thereby avoiding collision.

Compared with the artificial potential field method, the virtual structure method is more conducive to the organization of formation. This paper is designed to generate and transform the formation after the AUV reaching the specified depth, so the position and speed of the simplified formation reference point in the system are expressed as $p_r \in \mathbb{R}^2$, $\mathrm{u}_r \in \mathbb{R}^2$. We have

$$\begin{cases} p_{ri} = p_r + \mathcal{R}(\theta)\mathcal{L}r_i \\ \mathrm{u}_{ri} = \mathrm{u}_r + \dot{\mathcal{R}}(\theta)\mathcal{L}r_i + \mathcal{R}(\theta)\dot{\mathcal{L}}r_i \end{cases} \tag{15}$$

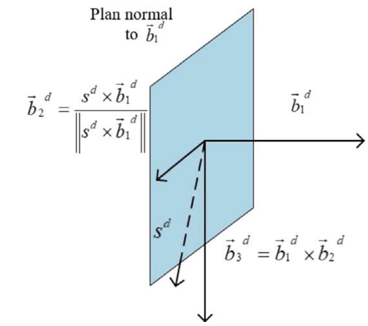

Fig. 3. Desired heading direction.

where $\mathcal{R}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ represents the formation rotation matrix of the virtual structure; $\mathcal{L} = \begin{bmatrix} l & 0 \\ 0 & l \end{bmatrix}$ is the formation zoom factor; r_i is the position of the ith mass point relative to the formation reference point, which represents the shape of the virtual structure. If r_i remains constant, the formation shape remains constant; if θ is expressed as the direction of the velocity vector of the formation reference point, the attitude of the formation is remain unchanged. Since the collision between AUVs is avoided by the artificial potential field, a step input is adopted for the change of the formation shape, that is, $r_i \to r_i'$. We avoid the design of $\dot{r}_i$, which simplifies the formation change.

4. Problem formulation

4.1. Trajectory generation

We choose an appropriate $s^d(t) \in S^2$, where S^2 is the unit sphere, to be transverse to the fixed thrust direction of the vehicle body which is $\vec{b}_{1_d}$.

The translational dynamics of the AUV is controlled by the total thrust $-fRe_1$. The total thrust f is directly controlled, and the direction of the total thrust $-Re_1$ is along the first body-fixed frame $-\vec{b}_1$. We will generate a sufficiently smooth and feasible trajectory by generating a set of given n waypoints. Each waypoint gives an inertial position $x^{d_i} \in \mathbb{R}^3$. For the given translation command $x^d(t)$, we choose the total thrust f and the desired direction $\vec{b}_{1_d}$ of the first body-fixed frame to stabilize translation dynamics. Once the desired direction of the first body-fixed frame $\vec{b}_{1_d}$ is chosen, one degree of freedom is left to choose to obtain the desired attitude $R_d = [\vec{b}_{1_d}, \vec{b}_{2_d}, \vec{b}_{3_d}] \in SO(3)$. We assume that $s^d(t)$ is not parallel to $\vec{b}_{1_d}$, so that we can get the desired attitude R_d:

$$\vec{b}_{2_d} = \frac{s^d \times \vec{b}_{1_d}}{\|s^d \times \vec{b}_{1_d}\|} \tag{16}$$

$$\vec{b}_{3_d} = \vec{b}_{1_d} \times \vec{b}_{2_d} \tag{17}$$

The control moment τ is chosen to follow this desired attitude. That is, we guarantee that $x(t) \to x^d(t)$ and $R(t) \to R_d(t)$ in finite time, the heading is defined as shown in Fig. 3.

4.2. Tracking error

The position error and velocity error of the tracking algorithm are given by:

$$\tilde{x} = x - x^d \tag{18}$$

$$\tilde{v} = v - v^d \tag{19}$$

Then we can know that

$$\dot{\tilde{x}} = \tilde{v} \tag{20}$$

$$\dot{\tilde{v}} = D_v(v)(\tilde{v} + v^d)/m - u^d - \dot{v}^d \tag{21}$$

where $u^d = fRe_1/m$. u^d can be considered as the control vector acting on the AUV, which is expressed in the inertial frame $\mathcal{I}$. The magnitude of the control vector $\|u^d\|$ will be the thrust magnitude f. The direction of the control vector is defined as $\bar{u}^d = u^d/\|u^d\|$.

We define R_d as the desired attitude, and Ω^d is the desired attitude angular velocity in the fixed body frame defined by R_d. The tracking errors of attitude and angular velocity evolve on the tangent bundle of the nonlinear space SO(3). Define the attitude and angular velocity tracking error as $\omega = \Omega - \Psi^T \Omega^d$. We choose the error function which on SO(3) is

$$\Psi(R, R_d) = R_d^T R \tag{22}$$

Take its derivative with respect to time:

$$\dot{\Psi} = \Psi\hat{\omega} \tag{23}$$

So the error dynamics turn into

$$\begin{aligned} J\dot{\omega} &= J(\hat{\omega}\Psi^T\Omega^d - \Psi^T\dot{\Omega}^d) \\ &\quad - (\widehat{\omega + \Psi^T\Omega})J(\omega + \Psi^T\Omega) + D_\Omega(\Omega)(\omega + \Psi^T\Omega^d) + \tau \end{aligned} \tag{24}$$

We solve the control problem in two steps. In the first step, we design u^d so that the tracking error $(\tilde{x}, \tilde{v})$ stabilizes to $(0, 0)$ within a finite-time. The thrust magnitude f is chosen as $f = m\|u^d\|$. We get the desired $R^d(t)$ so that the thrust direction of the fixed body $R^d e_1$ is aligned with the unit vector $\bar{u}^d$. In the second step, the attitude trajectory tracking controller is designed so that R can track the required attitude $R^d(t)$ in finite time.

5. Control theory

5.1. Finite-time stable translation control

Lemma 1. *Let g and h be non-negative real numbers and let $q \in (1, 2)$. Then*

$$g^{(\frac{1}{q})} + h^{(\frac{1}{q})} \geq (g + h)^{(\frac{1}{q})} \tag{25}$$

In addition, if both the base of the two terms on the left side of the inequality sign are non-zero, the above inequality is a strict inequality.

Theorem 1. *Define*

$$z(t) = \frac{\tilde{x}}{(\tilde{x}^T\tilde{x})^{1-\frac{1}{q}}} \tag{26}$$

We give the $u^d \in \mathbb{R}^3$ as

$$u^d = -\dot{v}^d + k\dot{z} + k\tilde{x} + D_v(v)(\tilde{v} + v^d)/m + \frac{kP(\tilde{v} + kz)}{[(\tilde{v} + kz)^T P(\tilde{v} + kz)]^{1-\frac{1}{q}}} \tag{27}$$

where

$$\dot{z} = \frac{(\tilde{x}^T\tilde{x})^{1-\frac{1}{q}}\tilde{v} - (2 - \frac{2}{q})(\tilde{x}^T\tilde{x})^{-\frac{1}{q}}(\tilde{x}^T\tilde{v})\tilde{x}}{(\tilde{x}^T\tilde{x})^{2-\frac{2}{q}}} \tag{28}$$

And $k > 0$, $P \in \mathbb{R}^{3\times3}$ is a positive definite control gain matrix. $\dot{\tilde{v}}$ stabilize the translational error dynamics in finite time. And $u^d(t) = f(t)Re_1/m$

Proof. For the desired translational motion, consider the following Lyapunov function:

$$V(\tilde{x}, \tilde{v}) = \frac{1}{2}k\tilde{x}^T\tilde{x} + \frac{1}{2}(\tilde{v} + kz)^T(\tilde{v} + kz), \tag{29}$$

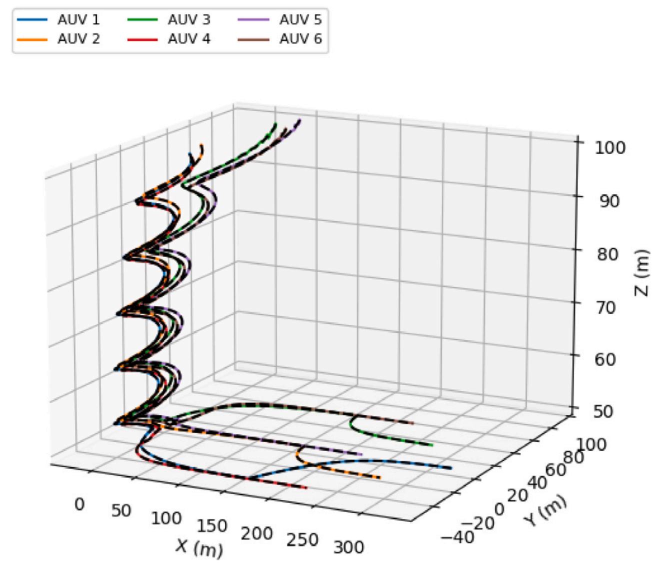

Fig. 4. Desired trajectories and the trajectories of algorithm realization.

where k is a positive scalar constant. We can see that

$$\begin{aligned} \dot{V} &= k\tilde{x}^T\tilde{v} + (\tilde{v} + kz)^T(\dot{\tilde{v}} + k\dot{z}) \\ &= k\tilde{x}^T\tilde{v} + (\tilde{v} + kz)^T(D_v(v)(\tilde{v} + v^d)/m - u^d - \dot{v}^d + k\dot{z}) \end{aligned} \tag{30}$$

We substitute Eq. (27) in the above equation, and after algebraic simplification, we get:

$$\begin{aligned} \dot{V} &= -k\{k(\tilde{x}^T\tilde{x})^{\frac{1}{q}} + [(\tilde{v} + kz)^T P(\tilde{v} + kz)]^{\frac{1}{q}}\} \\ &\leq -k(V)^{\frac{1}{q}} \end{aligned} \tag{31}$$

Among them, we use Lemma 1 and assume that the scalar k and eigenvalues of P are greater than 1. Therefore, according to Theorem 1 of Bhat and Bernstein (1995), we obtain the finite-time translation control at $(\tilde{x}, \tilde{v}) = (0, 0) \in \mathbb{R}^3$.

The control law for the magnitude of the force vector is:

$$f = m\|u^d\| \tag{32}$$

5.2. Generation of desired attitude trajectory

The direction of the control thrust given by $\bar{u}^d = u^d/\|u^d\|$ is regulated by the control attitude R.

In order to achieve stable tracking of the required translational motion and the required pointing direction, the attitude R must be controlled so that $Re_1 = \vec{b}_{1_d}$ is consistent with the direction $\bar{u}^d$ and s^d is transverse to Re_1. In order to achieve this, we construct an attitude trajectory R^d and give an attitude tracking scheme to ensure that $R(t) \to R^d(t)$ is in a finite-time.

From Eqs. (16) and (17), we can get $\vec{b}_{2_d}$ and $\vec{b}_{3_d}$. After this, we will obtain $R_d = [\vec{b}_{1_d}, \vec{b}_{2_d}, \vec{b}_{3_d}]$. There are several ways to design s^d. For example, in Section 6.1, we choose $s^d = e_3$ to get R_d based on waypoints design and AUV construction.

Our next step is to choose a control torque τ such that $R(t) \to R_d(t)$ in finite time. We will formulate the attitude tracking control scheme in Section 5.3.

5.3. Finite time stable attitude tracking control on TSO(3)

Lemma 2. *Let $K = diag(k_1, k_2, k_3)$, where $k_1 > k_2 > k_3 \geq 1$. Define*

$$s_K(\Psi) = \sum_{i=1}^{3} k_i(\Psi^T e_i) \times e_i \tag{33}$$

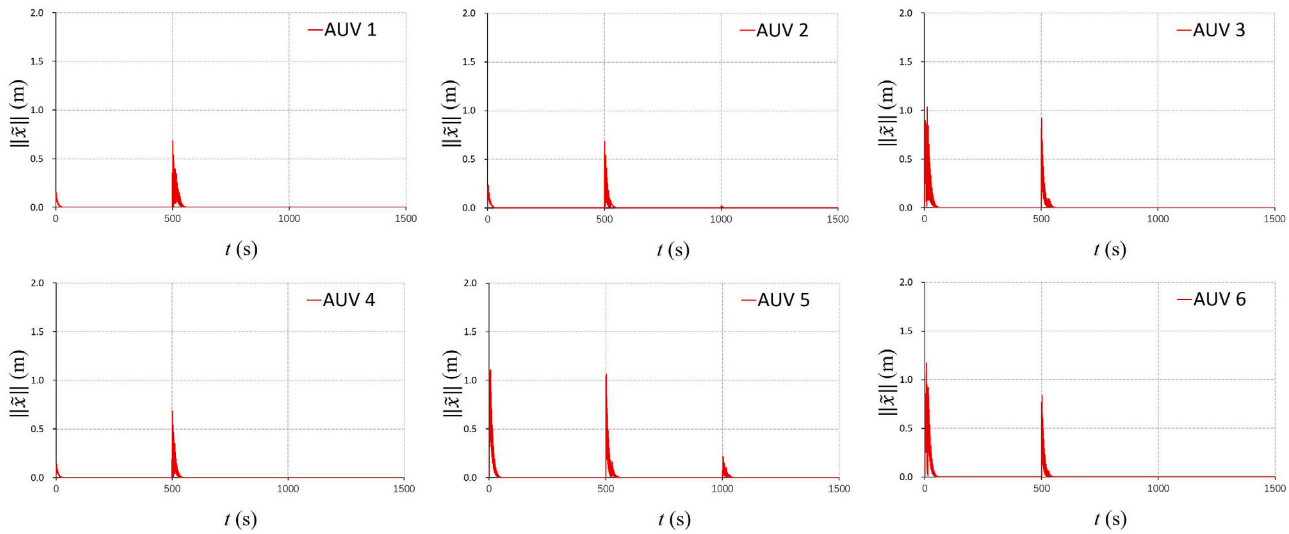

Fig. 5. Positions tracking errors.

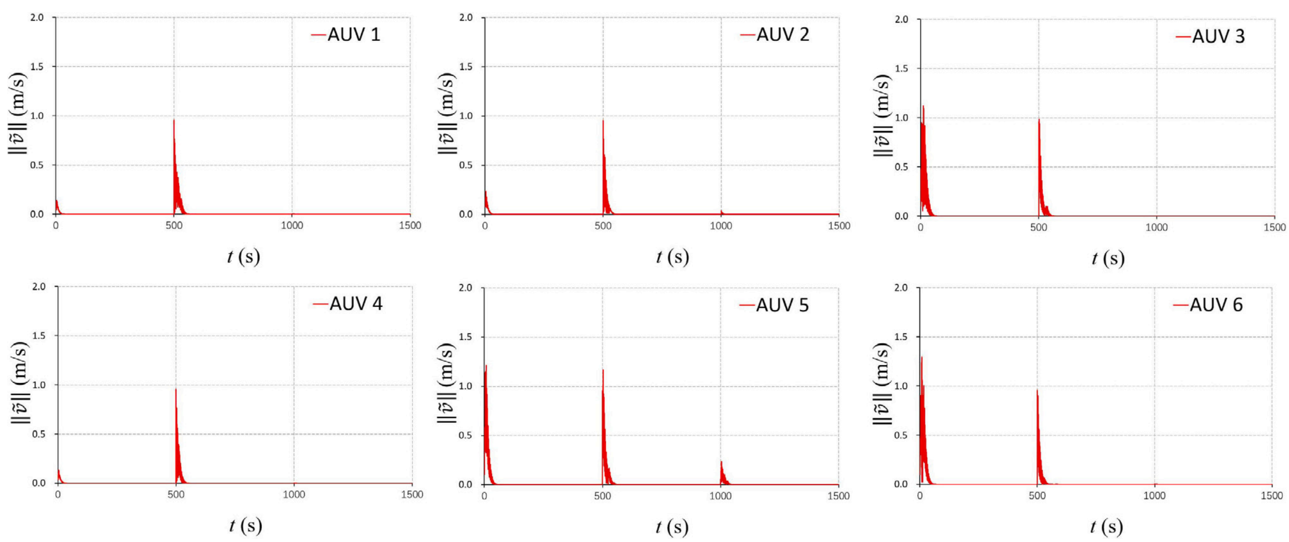

Fig. 6. Velocities tracking errors.

which makes $\frac{d}{dt}\langle K, I-\Psi\rangle = \omega^T s_K(\Psi)$. $\langle A, B\rangle = tr(A^T B)$, *which makes* $\langle K, I-\Psi\rangle$ *a Morse function defined on* $SO(3)$. *Let* $\mathcal{S} \subset SO(3)$ *be a closed subset that contains identity in its interior, defined as*

$$\begin{aligned}\mathcal{S} =&\{\Psi \in SO(3) : \Psi_{ii} \geq 0, \Psi_{ij}\Psi_{ji} \leq 0 \\ &\forall i, j \in \{1,2,3\} i \neq j\}\end{aligned} \tag{34}$$

For $\Psi \in \mathcal{S}$, *we can see*

$$s_K(\Psi)^T s_K(\Psi) \geq tr(K - K\Psi) \tag{35}$$

Proof. The proof of this lemma is given in Casau et al. (2013), and omitted here for brevity.

Theorem 2. *Consider the discretized rotational error kinematics and the dynamics of the AUV given in Eq. (24), where* $s_K(\Psi_k)$ *is defined. Define*

$$z_K(\Psi) = \frac{s_K(\Psi)}{(s_K^T(\Psi)s_K(\Psi))^{1-\frac{1}{p}}} \tag{36}$$

$$w(\Psi,\omega) = \frac{d}{dt}s_K(\Psi) = \sum_{i=1}^{3} k_i e_i \times (\omega \times \Psi^T e_i) \tag{37}$$

where p *is defined in Lemma 2. Let* L_Ω *be a positive definite control gain matrix, which make* $L_\Omega - J$ *be positive semi-definite, define* $k_p > 1$ *and* κ, *where*

$$\kappa^p = \frac{\sigma_{L,min}}{\sigma_{L,max}} > 0 \tag{38}$$

Define :

$$\Pi(\Psi,\omega) = \omega + \kappa z_k(\Psi), \tag{39}$$

$$H(y) = I - \frac{2(1-\frac{1}{p})}{y^T y} y y^T. \tag{40}$$

Then we give the feedback control law about τ*:*

$$\begin{aligned}\tau =& J[\Psi^T \dot{\Omega}^d - \frac{\kappa H s_k(\Psi)}{(s_k^T(\Psi)s_k(\Psi))^{1-\frac{1}{p}}} w(\Psi,\omega)] \\ &+ (\widehat{\Psi^T\Omega^d}) J(\Psi^T\Omega^d - \kappa z_K(\Psi)) + \kappa J(z_K(\Psi) \times \Psi^T \Omega^d) \\ &+ \kappa J(\omega + \Psi^T\Omega^d) \times z_K(\Psi) - k_p s_K(\Psi) \\ &- \frac{L_\Omega \Pi(\Psi,\omega)}{(\Pi^T(\Psi,\omega)L_\Omega\Pi(\Psi,\omega))^{1-\frac{1}{p}}} - D_\Omega(\Omega)(\omega + \Psi^T\Omega^d)\end{aligned} \tag{41}$$

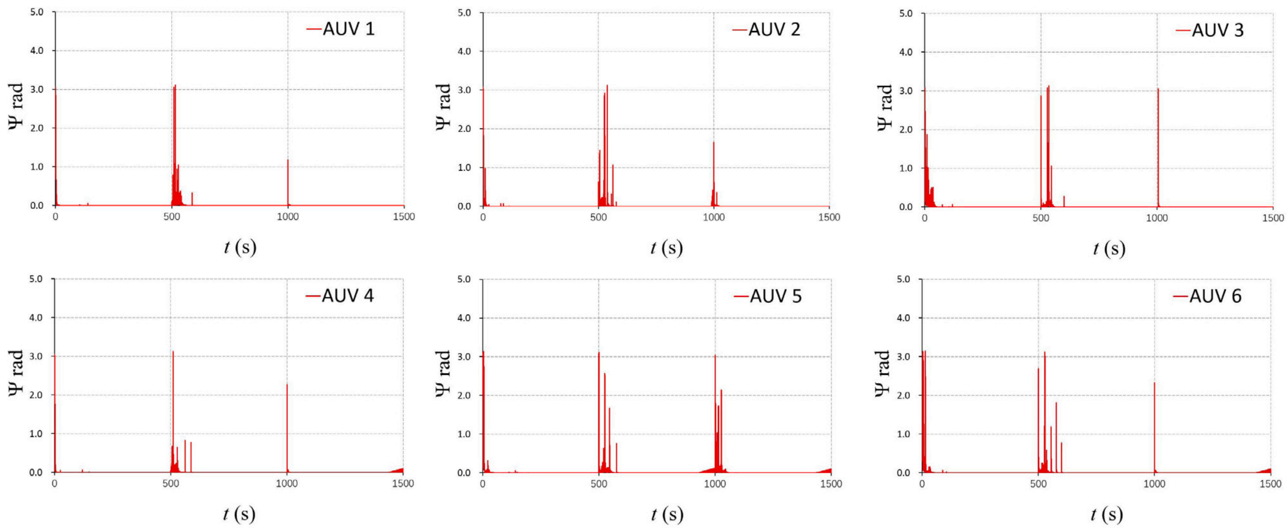

Fig. 7. Attitudes tracking errors.

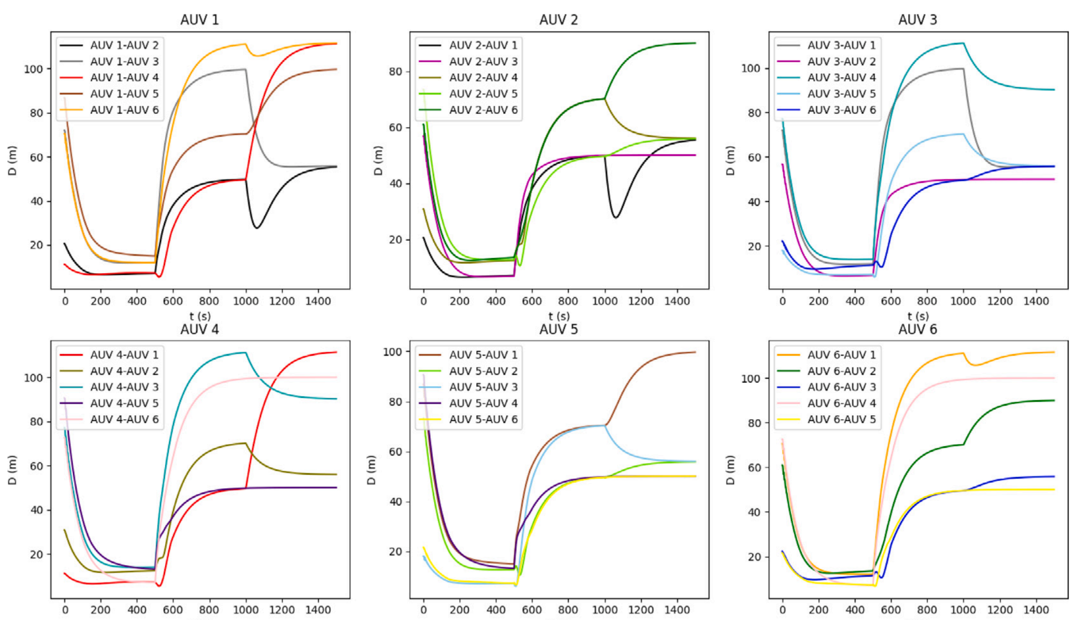

Fig. 8. Relative distance between AUVs in the formation.

As given in Eq. (24), the feedback attitude tracking error dynamics is stabilized to $(\Psi, \omega) = (I, 0)$ *in finite time.*

Proof. We consider the Lyapunov function

$$\mathcal{V}(\Psi,\omega) = k_p\langle K, I-\Psi\rangle + \Pi^T(\Psi,\omega)J\Pi(\Psi,\omega) \tag{42}$$

For the attitude dynamics of Eq. (24) with control law Eq. (41). The derivative of this Lyapunov function along the time of feedback dynamics is given by:

$$\begin{aligned}\dot{\mathcal{V}} =& k_p\omega^T s_K(\Psi) + \Pi^T(\Psi,\omega)J\dot{\Pi}(\Psi,\omega)\\ =& k_p\omega^T s_K(\Psi) + \Pi^T(\Psi,\omega)[\tau + J\Omega\times(\Psi^T\Omega^d - \kappa z_K(\Psi))\\ &+ J(\hat{\omega}\Psi^T\Omega^d - \Psi^T\dot{\Omega}^d) + D_\Omega(\Omega)(\omega+\Psi^T\Omega^d)\\ &+ \frac{J\kappa H s_k(\Psi)}{(s_k^T(\Psi)s_k(\Psi))^{1-\frac{1}{p}}}w(\Psi,\omega)]\end{aligned} \tag{43}$$

After substituting Eq. (41) into the above formula and performing several algebraic simplifications, we get

$$\begin{aligned}\dot{\mathcal{V}} =& -k_p\kappa(s_k^T(\Psi)s_k(\Psi))^{\frac{1}{p}} - (\Pi^T(\Psi,\omega)L_\Omega\Pi(\Psi,\omega))^{\frac{1}{p}}\\ \le& -\kappa(k_p\langle K, I-\Psi\rangle)^{\frac{1}{p}} - \kappa(\Pi^T(\Psi,\omega)J\Pi(\Psi,\omega))^{\frac{1}{p}}\end{aligned} \tag{44}$$

$(\Psi,\omega) \in S\times\mathbb{R}^3$, where $S \subset SO(3)$. Since the inequality Eq. (25), we can get

$$\begin{aligned}\dot{\mathcal{V}} \le& -\kappa(k_p\langle K, I-\Psi\rangle + \Pi^T(\Psi,\omega)J\Pi(\Psi,\omega))^{\frac{1}{p}}\\ =& -\kappa\mathcal{V}^{\frac{1}{p}}\end{aligned} \tag{45}$$

From Theorem 1 in Bhat and Bernstein (1995), it can be seen that the feedback attitude tracking control system is locally stable for a finite-time at $(\Psi,\omega) = (I,0)$. The attraction domain shown in the above analysis is $(\Psi,\omega) \in S\times\mathbb{R}^3$. The rest of the proofs that prove the almost global finite-time stability of attitude feedback control are the same as the proofs of similar results given in Bohn and Sanyal (2016), and are omitted here.

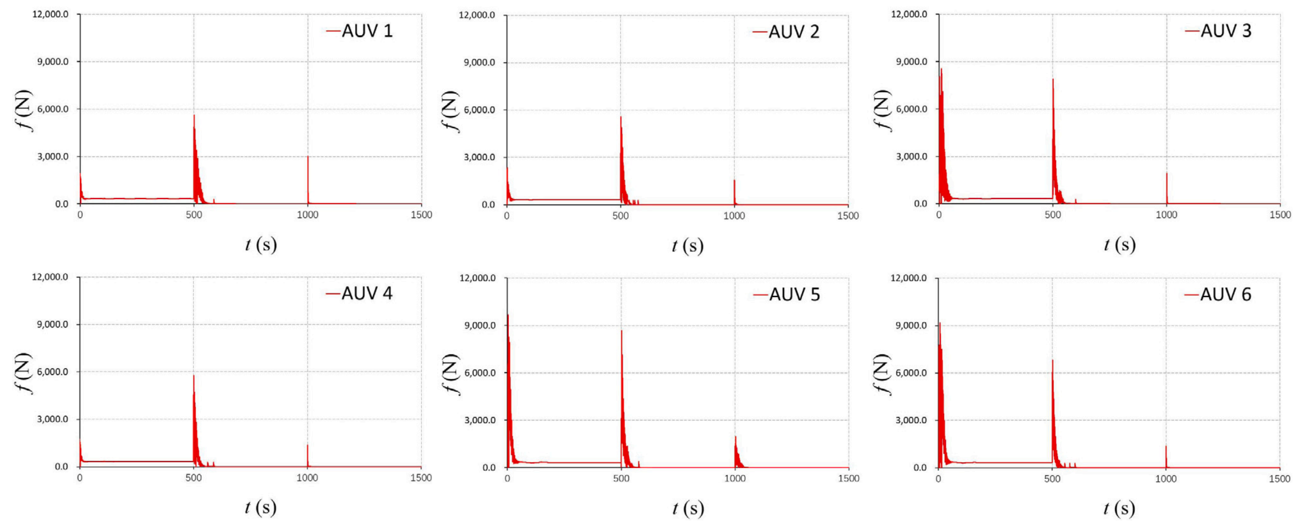

Fig. 9. Norms of the control torques.

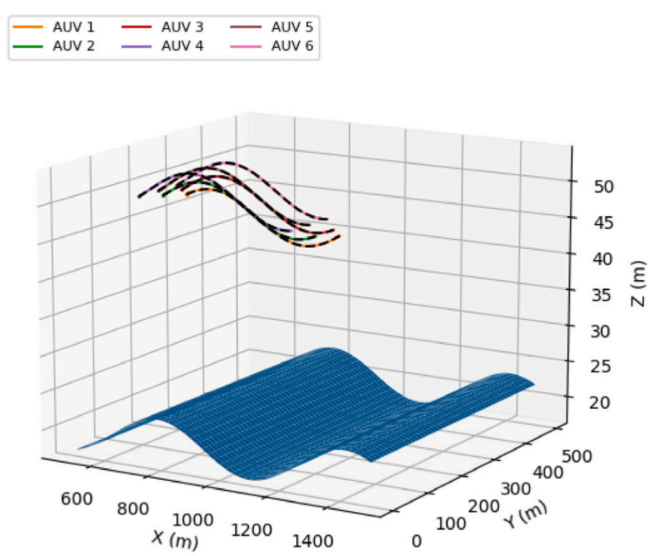

Fig. 10. Desired trajectories and the trajectories of algorithm realization.

5.4. Stability of the feedback system on TSE(3)

Theorem 3. *For the generated state trajectories* $(x^d(t), R^d(t), \upsilon^d(t), \Omega^d(t)) \subset TSE(3)$, *the overall feedback control system is stable for a finite-time, its convergence region is almost global in the state space.*

Proof. The attitude tracking control scheme given in Theorem 2 ensures that $R(t) = R^d(t)$ for time $t \geq T_1$ where $T_1 > 0$ is finite. Therefore, for $t \geq T_1$, we can obtain $Re_1 = r_1^d = \bar{u}^d$. From (32), $fR(t)e_1 = mu^d(t)$, $t \geq T_1$ where $u^d(t)$ is given in Eq. (27). Theorem 1 ensures that there exists a finite-time $T_2 > 0$ such that $(x(t), \upsilon(t)) = (x^d(t), \upsilon^d(t))$ for times $t \geq T_1 + T_2$.

6. Simulation results

The continuous motion equation is discretized in the form of a Lie group variational integrator, which retains the structure of the configuration space, that is, the Lie group SE(3). The obtained Lie group variational integrator is similar as that used in Sanyal et al. (2010) and Viswanathan et al. (2018), which is:

$$\begin{cases} R_{k+1} = R_k F_k, \\ x_{k+1} = \mathbb{h} R_k \upsilon_k + x_k, \\ M\upsilon_{k+1} = MF_k^T \upsilon_k + \mathbb{h} D_\upsilon(\upsilon_k) - \mathbb{h} f_k e_1, \\ \hat{\Omega}_{k+1}^d = log(R_k^{d^T} R_{k+1}^d)/\mathbb{h}, \\ J\Omega_{k+1} = F_k^T J\Omega_k + \mathbb{h} D_{\Omega_k}\Omega_k + \mathbb{h}\tau, \end{cases} \tag{46}$$

where $\mathbb{h} = \Delta t$ and $F_k \approx exp(h\hat{\Omega}_k) \in SO(3)$ guarantees that R_k evolves on SO(3).

This section introduces the numerical simulation results. The time step $\Delta t = 0.01$ s. For simulation purposes, vehicle parameters are selected (more information see Healey and Lienard (1993)):

$m = 5454.54$ kg,

$M_f = diag[-7.6e{-3}, -5.5e{-2}, -2.4e{-1}]$

$J = diag[2037.999, 13\,586.983, 13\,586.9966]$ kg m^2

The drag matrices are $D_\upsilon(\upsilon) = -[0; 0.1; 0.3]$ kg s^{-1} and $D_\Omega(\Omega) = -[0.011; 0; 0.016]$ s^{-1}. The control gain matrices are taken to be $P = 3.5I$ and $L_\Omega = 3.5I$. The simulations begin at time $t = 0$ and the initial conditions are

$\upsilon(0) = [0, 0, 0]^T$ m/s, $R(0) = I$, $\Omega(0) = [0, 0, 0]^T$ rad/s.

6.1. Simulation of the spiral descent and the formation and transformation of formations of the AUVs

We simulated 6 AUVs randomly deployed within a water surface with a diameter of $D = 100$ m. The time period is the time for the underwater vehicles from releasing to the specified depth $H = 50$ m and form the formation and transform the formation. The communication between AUVs selects a fixed time period and a random time offset communication method. On the premise of 6 AUVs, while the AUVs are descending in a spiral, the artificial potential field method is introduced to avoid obstacles within the team to generate a safe distance. And after the AUVs reaching the specified depth, the forming of the formation and transformation were performed to complete the task of detection.

Fig. 4 depicts the feasibility of the proposed scheme. Figs. 5 and 6 show the error of the position and velocity of the six AUVs from the expected position and speed. It can be seen that these errors converge in a finite-time. Fig. 7 shows the error between the attitude of the six AUVs and the desired attitude. Since the introduction of artificial potential field force is considered as interference force, this will cause the AUVs to produce attitude errors when avoiding obstacles within the

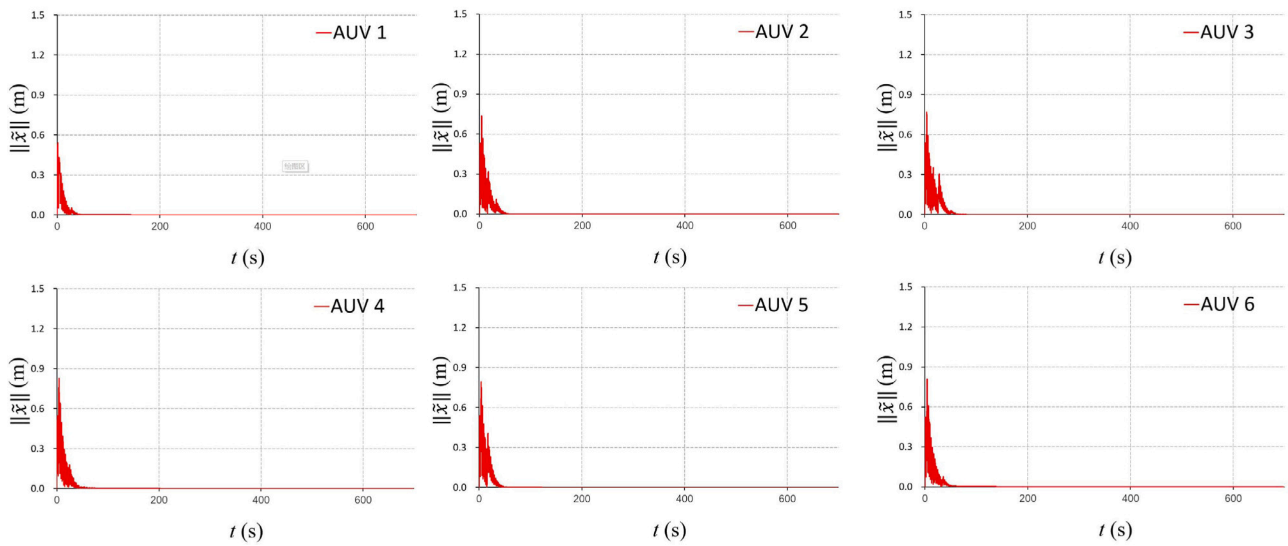

Fig. 11. Positions tracking errors.

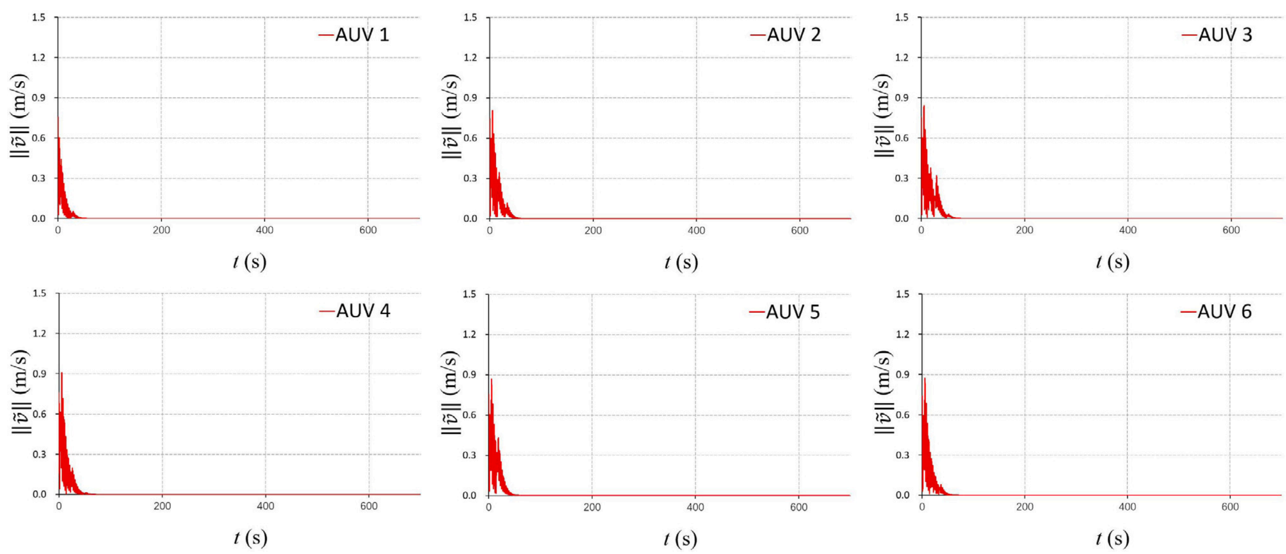

Fig. 12. Velocities tracking errors.

team, forming formations and changing formations, and the errors will converge within a finite-time. Fig. 8 shows the distance D between each AUV and other AUVs in the simulation process. During the descending process, in order to avoid collision between AUVs, the safety distance is set to 10 m. After that, the cooperative system starts to perform detection tasks and forms and changes the formation after reaching the desired depth. Fig. 8 shows the stability of the entire coordination system after the formation is maintained and the formation is changed. Fig. 9 shows the norms of the control torques.

6.2. The simulation of AUVs' attitudes adjustment according to the change of seabed topography

Our second simulated section is after the first section, when the AUVs reaches the design depth and start to search the seabed. The communication between AUVs selects a fixed time period and a random time offset communication method. In order to get better detection of the seabed, the AUVs can adjust their attitudes according to the vertical downward sensor. The vertical downward sensor can detect the changes of the distance from the AUVs to the seabed, and according to this change, the derivative of the change of the seabed topography in the e_3 direction can be obtained. After that, the AUVs can make corresponding attitude changes based on the obtained data, so that the forward direction of the AUVs and the seabed topography are kept relatively parallel, so that better seabed detection results can be obtained. The simulated seabed topography is:

$$Z(X) = 3sin(0.01X) + 20$$

where $X \in [500, 1500]$, $Y \in [1, 500]$. Fig. 10 depicts the feasibility of the proposed scheme. Figs. 11 and 12 show the error of the position and velocity of the six AUVs from the expected position and speed. It can be seen that these errors converge in a finite-time. Fig. 13 shows the error between the attitude of the six AUVs and the desired attitude. Fig. 14 shows the distance D between each AUV and other AUVs in the simulation process and the stability of the entire coordination system after the formation is formed. Fig. 15 shows the norms of the error $\tilde{H}$ between the actual trajectories of each AUV and the expected height from the seabed. Fig. 16 shows the norms of the control torques.

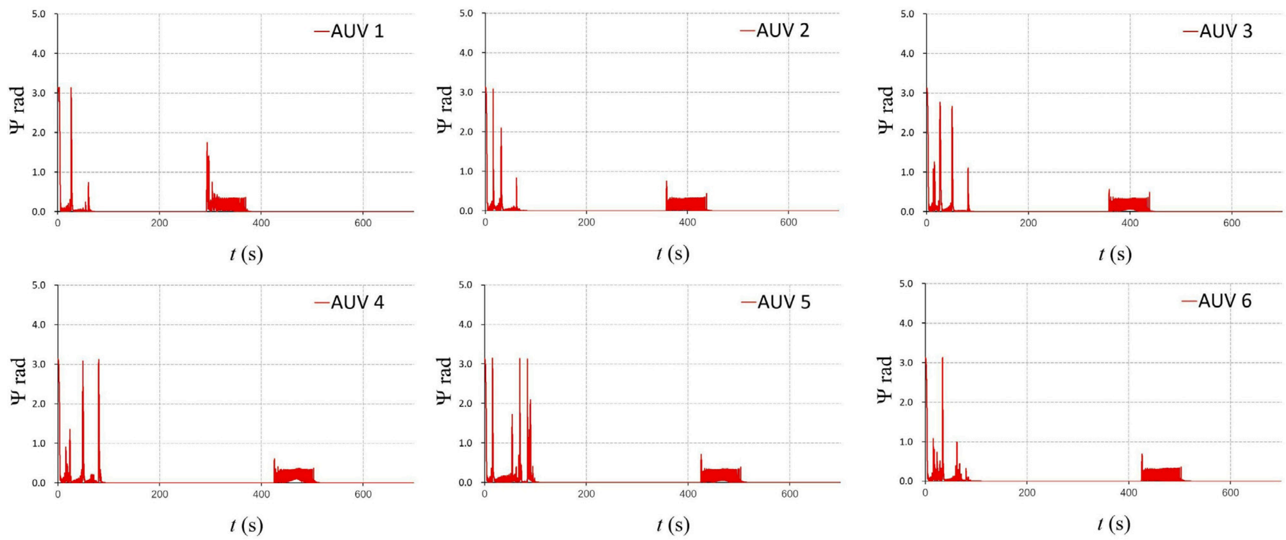

Fig. 13. Attitudes tracking errors.

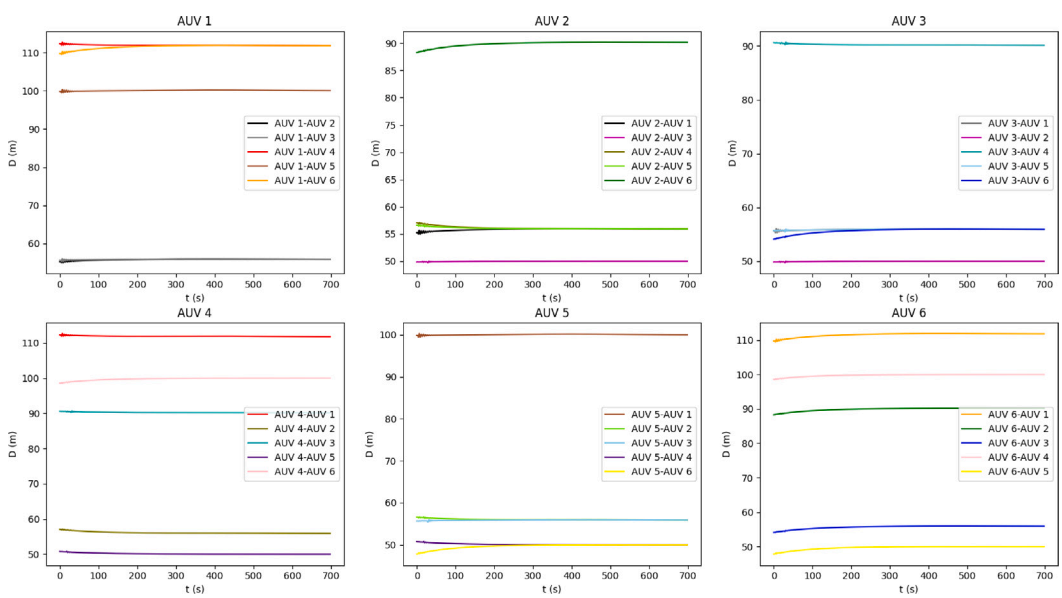

Fig. 14. Relative distance between AUVs in the formation.

7. Conclusion

This essay presents a finite-time stable position and attitude tracking control method based on the combination of artificial potential field and virtual structure for the formation control of multiple AUVs. We give a control vector to make the AUV reach the desired trajectory in a finite-time, and then we use the control vector and the desired pointing direction to generate the desired attitude. We give a finite-time attitude tracking method to track the desired attitude. After that, we use the artificial potential field and the desired trajectory generated by the virtual structure to verify the effectiveness of the method. Finally, we simulate the deployment of the AUVs until they reach the desired depth and the form the formation and do the change of formation. We simulate the changes in the attitudes of the AUVs in response to the changes in the seabed topography during depth detection. These simulations verify the effectiveness of the method. In the future, we will focus on the disturbance caused by the artificial potential field force which makes the formation to improve the anti-jamming performance of the AUVs.

CRediT authorship contribution statement

Qingzhe Zhen: Investigation, Conceptualization, Methodology, Formula analysis, Data curation, Writing – original draft, Writing – review & editing, Validation, Visualization. **Lei Wan:** Review & editing, Supervision, Validation. **Yulong Li:** Validation, Writing, Formula analysis, Resources. **Dapeng Jiang:** Review & editing, Supervision, Validation, Funding acquisition, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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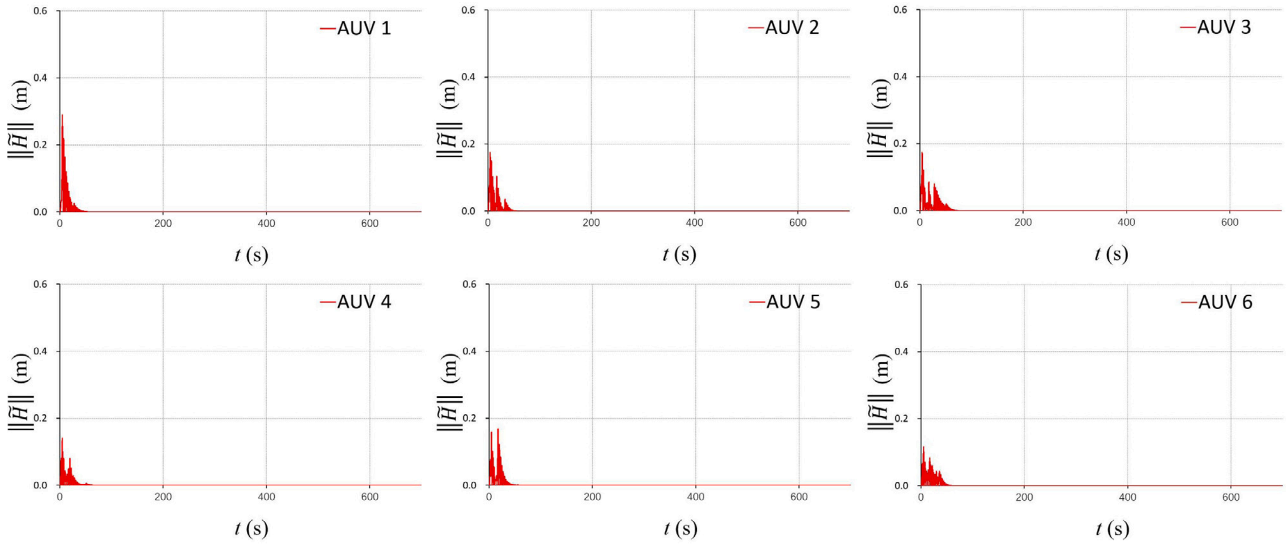

Fig. 15. Height tracking errors.

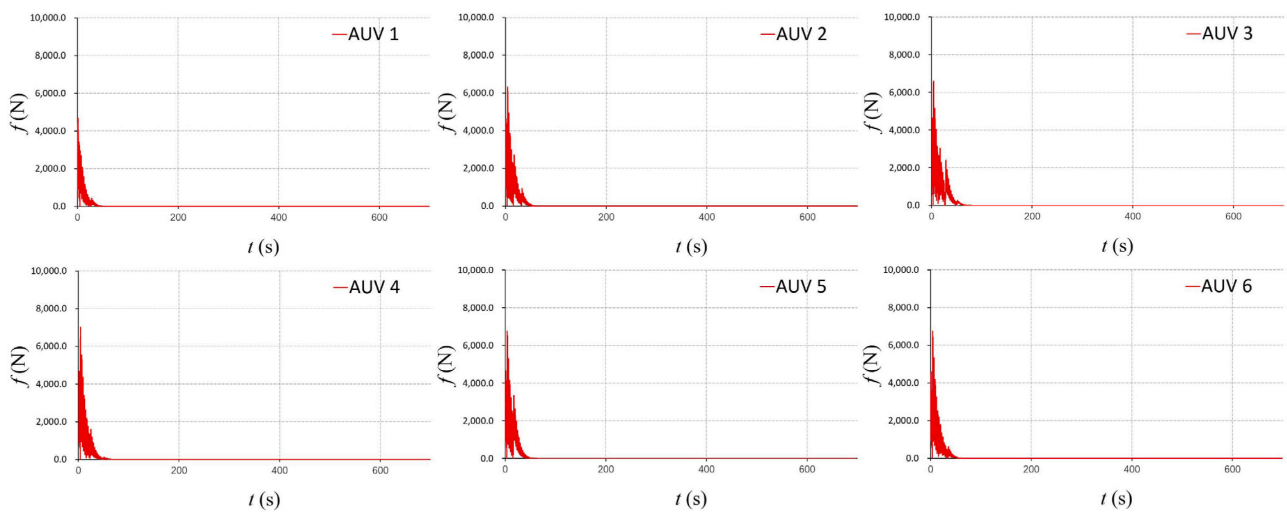

Fig. 16. Norms of the control torques.

References

Bhat, S.P., Bernstein, D.S., 1995. Lyapunov analysis of finite-time differential equations. In: Proceedings of 1995 American Control Conference-ACC'95, Vol. 3. IEEE, pp. 1831–1832.

Bohn, J., Sanyal, A.K., 2016. Almost global finite-time stabilization of rigid body attitude dynamics using rotation matrices. Internat. J. Robust Nonlinear Control 26 (9), 2008–2022.

Cai, L., Sun, Q., Xu, T., Ma, Y., Chen, Z., 2020. Multi-AUV collaborative target recognition based on transfer-reinforcement learning. IEEE Access 8, 39273–39284.

Cao, X., Sun, H., Jan, G.E., 2018. Multi-AUV cooperative target search and tracking in unknown underwater environment. Ocean Eng. 150, 1–11.

Cao, X., Zhu, D., Yang, S.X., 2015. Multi-AUV target search based on bioinspired neurodynamics model in 3-D underwater environments. IEEE Trans. Neural Netw. Learn. Syst. 27 (11), 2364–2374.

Casau, P., Sanfelice, R.G., Cunha, R., Cabecinhas, D., Silvestre, C., 2013. Global trajectory tracking for a class of underactuated vehicles. In: 2013 American Control Conference. IEEE, pp. 419–424.

Chang, A.Y., Chen, C.-F., Chiu, L.Y., Liu, E., Chiu, C.-S., Reeder, D.B., 2014. Three-dimensional acoustic propagation effect in subaqueous sand dune field. J. Acoust. Soc. Am. 136 (4), 2178.

Chaturvedi, N.A., McClamroch, N.H., 2006. Almost global attitude stabilization of an orbiting satellite including gravity gradient and control saturation effects. In: 2006 American Control Conference. IEEE, pp. 6–pp.

Chen, Y.-L., Ma, X.-W., Bai, G.-Q., Sha, Y., Liu, J., 2020. Multi-autonomous underwater vehicle formation control and cluster search using a fusion control strategy at complex underwater environment. Ocean Eng. 216, 108048.

Dai, L., Zheng-Lin, L., Ruo-Yun, L., 2021. Sound propagation in shallow water with periodic rough bottom. Acta Phys. Sin. 70 (3).

Do, K.D., 2007. Bounded controllers for formation stabilization of mobile agents with limited sensing ranges. IEEE Trans. Automat. Control 52 (3), 569–576.

Fernandes, P.G., Stevenson, P., Brierley, A.S., Armstrong, F., Simmonds, E.J., 2003. Autonomous underwater vehicles: future platforms for fisheries acoustics. ICES J. Mar. Sci. 60 (3), 684–691.

Fiorelli, E., Leonard, N.E., Bhatta, P., Paley, D.A., Bachmayer, R., Fratantoni, D.M., 2006. Multi-AUV control and adaptive sampling in Monterey bay. IEEE J. Ocean. Eng. 31 (4), 935–948.

Fossen, T.I., 2011. Handbook of Marine Craft Hydrodynamics and Motion Control. John Wiley & Sons.

Gazi, V., Fidan, B., Ordonez, R., İlter Köksal, M., 2012. A target tracking approach for nonholonomic agents based on artificial potentials and sliding mode control. J. Dyn. Syst. Meas. Control 134 (6), 061004.

Ge, H., Chen, G., Xu, G., 2018. Multi-AUV cooperative target hunting based on improved potential field in a surface-water environment. Appl. Sci. 8 (6), 973.

Healey, A.J., Lienard, D., 1993. Multivariable sliding mode control for autonomous diving and steering of unmanned underwater vehicles. IEEE J. Ocean. Eng. 18 (3), 327–339.

Leonard, N.E., Fiorelli, E., 2001. Virtual leaders, artificial potentials and coordinated control of groups. In: Proceedings of the 40th IEEE Conference on Decision and Control (Cat. No. 01CH37228), Vol. 3. IEEE, pp. 2968–2973.

Sanyal, A., Chaturvedi, N., 2008. Almost global robust attitude tracking control of spacecraft in gravity. In: AIAA Guidance, Navigation and Control Conference and Exhibit. p. 6979.

Sanyal, A., Nordkvist, N., Chyba, M., 2010. An almost global tracking control scheme for maneuverable autonomous vehicles and its discretization. IEEE Trans. Automat. Control 56 (2), 457–462.

Singh, S., Sanyal, A., Smith, R., Nordkvist, N., Chyba, M., 2009. Robust tracking control of autonomous underwater vehicles in the presence of disturbance inputs. In: International Conference on Offshore Mechanics and Arctic Engineering, Vol. 43444. pp. 345–351.

Tanner, H.G., Kumar, A., 2005. Towards decentralization of multi-robot navigation functions. In: Proceedings of the 2005 IEEE International Conference on Robotics and Automation. IEEE, pp. 4132–4137.

Urick, R.J., 1954. The backscattering of sound from a harbor bottom. J. Acoust. Soc. Am. 26 (2), 231–235.

Viswanathan, S.P., Sanyal, A.K., Samiei, E., 2018. Integrated guidance and feedback control of underactuated robotics system in SE (3). J. Intell. Robot. Syst. 89 (1), 251–263.

Wen, L., Zheng-Lin, L., Ren-He, Z., Ji-Xing, Q., Jun, L., Ming-Xing, N., 2015. The effects of seamounts on sound propagation in deep water. Chin. Phys. Lett. 32 (6), 064302.

Woolsey, C.A., 2006. Directional control of a slender, underactuated AUV using potential shaping. In: Proceedings of the 45th IEEE Conference on Decision and Control. IEEE, pp. 6826–6831.

Wu, J., Bingham, R.C., Ting, S., Yager, K., Wood, Z.J., Gambin, T., Clark, C.M., 2019. Multi-AUV motion planning for archeological site mapping and photogrammetric reconstruction. J. Field Robotics 36 (7), 1250–1269.

Wynn, R.B., Huvenne, V.A., Le Bas, T.P., Murton, B.J., Connelly, D.P., Bett, B.J., Ruhl, H.A., Morris, K.J., Peakall, J., Parsons, D.R., et al., 2014. Autonomous underwater vehicles (AUVs): Their past, present and future contributions to the advancement of marine geoscience. Mar. Geol. 352, 451–468.

Xia, G., Zhang, Y., Zhang, W., Chen, X., Yang, H., 2021. Multi-time-scale 3-D coordinated formation control for multi-underactuated AUV with uncertainties: Design and stability analysis using singular perturbation methods. Ocean Eng. 230, 109053.

Xing, W., Zhao, Y., Karimi, H.R., 2017. Convergence analysis on multi-AUV systems with leader-follower architecture. IEEE Access 5, 853–868.

Yan, Z., Yang, Z., Yue, L., Wang, L., Jia, H., Zhou, J., 2019. Discrete-time coordinated control of leader-following multiple auvs under switching topologies and communication delays. Ocean Eng. 172, 361–372.

Zhu, D., Liu, Y., Sun, B., 2018. Task assignment and path planning of a multi-AUV system based on a glasius bio-inspired self-organising map algorithm. J. Navig. 71 (2), 482–496.